CITY
ST GEORGE'S
UNIVERSITY OF LONDON

School of Science & Technology

IN1011 Operating Systems
Stage 1 Mock Examination B

May/June

Examination duration: **90** minutes

Division of marks:	Answer **3** of the **4** questions (20 marks for each question). If more than **3** questions are answered, the best **3** will be counted. Total mark is out of **60**
Other instructions:	All necessary working must be shown.

BEGIN EACH QUESTION ON A FRESH PAGE

Number of answer books to be provided: 1
Calculators permitted: Yes
Dictionaries permitted: None
Can question paper be removed from the examination room: No

Additional materials: **None**

Question 1

a. Describe 5 distinct scenarios that each illustrate how the OS relies on interrupts to perform its activities. Each scenario should clearly state how the interrupt arises (i.e. the situation/event that makes the interrupt necessary), the hardware that triggers the interrupt signal, and what the OS is expected to do in response to the interrupt. **[15 Marks]**

b. Let $\boldsymbol{x}$ be a positive integer (e.g. 1, 2, 3, etc.). Furthermore, let $\boldsymbol{x1}$ and $\boldsymbol{1x}$ be 2-digit numbers in base $\boldsymbol{x}$+1. What is $\boldsymbol{x1} + \boldsymbol{1x}$ in base $\boldsymbol{x}$+1? Show how you arrived at your answer using the definition of what it means to write numbers in base $\boldsymbol{x}$+1.
(*hint: write* $\boldsymbol{x1}$ *in powers of* $\boldsymbol{x}$*+1, write* $\boldsymbol{1x}$ *in powers of* $\boldsymbol{x}$*+1, then add these together*) **[2 Marks]**

c. The command "**`chmod 522 mybinaryfile`**" is executed in a Bash terminal.

   i. In what base is the 3-digit number **`522`** written in? **[1 Mark]**

   ii. How many distinct permission settings can be carried out using such 3-digit numbers with **`chmod`**? How did you deduce your answer? **[2 Marks]**

Question 2

a. Consider what happens when the following program G() is executed.

```
G( )
{
    int c = 0;
    int id = fork();

    if( id==0 ){ c = c+1; }
    else{
        id=fork();
        c=c+1;

        if( id==0 ){ c = c+1; }
     }
}
```

i. How many processes are there in total? **[1 Mark]**

ii. How many parent processes? **[1 Mark]**

iii. How many child processes? **[1 Mark]**

iv. What are the different final values of c? **[3 Marks]**

v. What final value of c is associated with the final process created? **[2 Marks]**

vi. What final value of c is associated with the parent of the process in the previous question v? **[2 Marks]**

vii. Why must G( ) cause interrupts during its execution? **[1 Mark]**

b. What is *the memory hierarchy*? Your answer should clearly indicate 3 ways the hierarchy constrains modern computers. **[4 Marks]**

c. What is *the principle of spatial locality*? **[2 Marks]**

d. Using an example involving computer hardware resources, illustrate how *the principle of spatial locality* can be used to partly overcome *memory hierarchy* constraints. **[3 Marks]**

Question 3

a. What is the *Effective Access Time* (EAT) in a demand-paged system? Explain.

[4 Marks]

b. Consider a system in which 100% of page-faults require a page to be swapped in from disk and 50% of page-faults require a victim page to be swapped out. If the page size is 4KiB and disk read and write speeds are both 300 MiB per second, calculate the *average page-fault service time* (assume all overheads apart from swap time are negligible). Give your answer to two significant figures.

[6 Marks]

c. For a system with page size 4,096 bytes and a process with 123,456 bytes, calculate:
   i. number of pages used, **[2 Marks]**
   ii. internal fragmentation. **[2 Marks]**

d. In a virtual memory system based on paging, what is a page table? Draw a diagram to show how a page table (without any translation look-aside buffer) is used to translate a virtual memory address to a physical memory address.

[6 Marks]

Question 4

a. A process that requires 8.2 MB of space is swapped to a hard disk. The transfer rate to the disk is 108.4 MB/s and there is a further latency of 7.3 microseconds. Calculate the total Context Switch Swapping time (that is swap out and swap in) of this process. Write down your answer in milliseconds.

[5 Marks]

b. Describe the way files are organised in a single-level directory. What are the main two problems of this structure? Describe another directory structure that addresses the problems of the single-level directory.

[7 Marks]

c. A group of files that exists in a directory have the permissions shown in the image below. Identify the permissions in a numeric way.

```
-rw--wx--x
--wxr-x-w-
-------rwx
```

[3 Marks]

d. In a demand-paged system, what is a *page-fault*? Outline the procedure by which a page-fault is handled.

[5 Marks]

End of Exam